Supplementary simulation results

Task-Specified Path–Speed Preference Control for an Autonomous Single-Track Robot

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Reading the evidence

This supplement gives the route-level data omitted from the focused article. All scores use the same original world-frame path and destination, a 10 s observation window, and the frozen task contract. "Goal" means entry into the original 1 m destination disk; H10 means upright survival throughout. A narrow pass additionally needs a buffered path corridor and 8–10 s original-path return. A wide pass adds an actual mean turn speed of at least 85% of the request. The 0.5 m circular buffer is an anchor-based vehicle-envelope proxy. Counts by route use two paired prepared states, not independent policy-training seeds.

U170 is one frozen preference-conditioned middle network. Each $\alpha \in \{0, 0.25, 0.5, 0.75, 1\}$ is an input to the same weights. H75 is a separate earlier checkpoint. The geometric rule uses no middle-network output; all matched rows share the lower balance stack and 10 Hz causal path reconnection. The new nominal task combinations were declared before execution. Signed pulses, physical-domain extrema and 90° routes were added after earlier outcomes had been inspected and are exploratory. Physical-domain and nominal panels have different prepared states and must be read separately.

1 Training review

The U170 continuation record spans updates 53–202. Its sampled training reward changes with the rollout distribution and is not a fixed-task performance curve. The source log records 150 updates, 2376 accepted minibatches, zero physical failures, 119 domain exits and three target-KL-shortened updates (155, 175 and 188). The frozen checkpoint is assessed by the executed task panels below. Figure 1 shows the saved reward, domain-exit, KL and critic-loss trends.

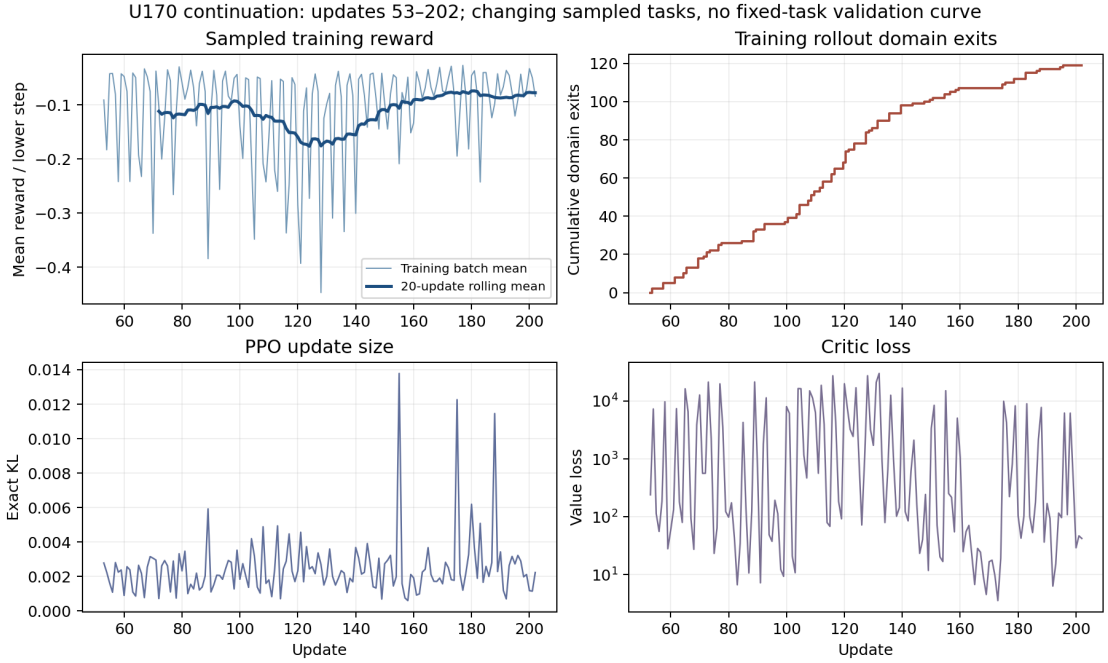


Figure 1: Sampled training diagnostics from the archived U170 continuation.

Table 1: U170 training windows; reward values are mean per sampled step.

Window	Updates	Reward	SD	Domain exits	Exact KL
Early	53–92	-0.1075	0.0890	33	0.0021
Middle	108–147	-0.1458	0.1154	53	0.0025
Late	163–202	-0.0763	0.0457	12	0.0029

2 New nominal task combinations

Four exact low/high single/S route–speed combinations were run from prepared banks 2 and 5. Table 2 reports every route and preference for both H75 and U170. Each row has two physical episodes; all 80 episodes reached the original goal and survived H10. The task completion columns add the path, return and speed tests stated above. Turn speed is the true vehicle-forward speed averaged over the declared turn interval. "Max cross" is maximum absolute original-path cross-track error before first goal entry.

Table 2: Nominal task outcomes by frozen checkpoint, route and preference.

Model	Route	α	Narrow	Wide	Speed	Max cross
H75	High S L	0	2/2	2/2	2.781	0.247
H75	High S L	0.25	1/2	2/2	2.836	0.403
H75	High S L	0.5	0/2	2/2	2.903	0.530
H75	High S L	0.75	0/2	2/2	2.981	0.658
H75	High S L	1	0/2	1/2	3.050	0.906
H75	High single R	0	2/2	2/2	2.667	0.300
H75	High single R	0.25	0/2	2/2	2.732	0.521
H75	High single R	0.5	0/2	2/2	2.829	0.697
H75	High single R	0.75	0/2	2/2	2.941	0.852
H75	High single R	1	0/2	0/2	3.019	0.990
H75	Low S R	0	2/2	2/2	1.734	0.180
H75	Low S R	0.25	2/2	2/2	1.774	0.185
H75	Low S R	0.5	2/2	2/2	1.840	0.187
H75	Low S R	0.75	2/2	2/2	1.896	0.278
H75	Low S R	1	1/2	2/2	1.933	0.404
H75	Low single L	0	2/2	1/2	1.619	0.164
H75	Low single L	0.25	2/2	2/2	1.671	0.158
H75	Low single L	0.5	2/2	2/2	1.729	0.150
H75	Low single L	0.75	2/2	2/2	1.785	0.139
H75	Low single L	1	2/2	2/2	1.829	0.162
U170	High S L	0	0/2	2/2	2.694	0.449
U170	High S L	0.25	0/2	2/2	2.741	0.516
U170	High S L	0.5	0/2	2/2	2.790	0.594
U170	High S L	0.75	0/2	2/2	2.830	0.660
U170	High S L	1	0/2	2/2	2.887	0.720
U170	High single R	0	2/2	0/2	2.507	0.211
U170	High single R	0.25	2/2	0/2	2.544	0.364
U170	High single R	0.5	0/2	2/2	2.602	0.557
U170	High single R	0.75	0/2	2/2	2.697	0.715
U170	High single R	1	0/2	2/2	2.807	0.850
U170	Low S R	0	2/2	0/2	1.616	0.216
U170	Low S R	0.25	2/2	0/2	1.647	0.229

Model	Route	α	Narrow	Wide	Speed	Max cross
U170	Low S R	0.5	2/2	0/2	1.682	0.242
U170	Low S R	0.75	2/2	2/2	1.719	0.256
U170	Low S R	1	2/2	2/2	1.762	0.278
U170	Low single L	0	2/2	0/2	1.558	0.215
U170	Low single L	0.25	2/2	0/2	1.593	0.227
U170	Low single L	0.5	2/2	2/2	1.637	0.241
U170	Low single L	0.75	2/2	2/2	1.691	0.259
U170	Low single L	1	2/2	2/2	1.744	0.280

3 Signed-force pulse extension

A world-fixed signed 36 N force was applied at 2.8–2.95 s on the same four routes and banks. Table 3 groups the full 48-case matrix by route, sign and method (two states per row). All 48 remained upright. The positive example in the article is the +36 N low S right and high single right subset. Table 4 exposes those four matched route–state contexts without aggregating away speed, path deviation, goal entry or the 8–10 s return error. The subset was identified after viewing the full matrix; the negative sign and other routes remain in Table 3.

Table 3: Full signed-pulse matrix grouped over two prepared states.

Route	Force	Method	Goal	Narrow	Wide	Speed	Max cross
High S L	-36	Rule	2/2	0/2	0/2	3.038	3.057
High S L	-36	U170 $\alpha = 0$	2/2	0/2	0/2	2.784	2.386
High S L	-36	U170 $\alpha = 1$	0/2	0/2	0/2	3.056	2.566
High S L	+36	Rule	2/2	0/2	2/2	3.034	0.664
High S L	+36	U170 $\alpha = 0$	2/2	0/2	2/2	2.657	0.852
High S L	+36	U170 $\alpha = 1$	0/2	0/2	0/2	3.005	2.376
High single R	-36	Rule	2/2	0/2	0/2	3.022	0.541
High single R	-36	U170 $\alpha = 0$	2/2	0/2	0/2	2.526	0.506
High single R	-36	U170 $\alpha = 1$	2/2	0/2	0/2	2.933	1.189
High single R	+36	Rule	2/2	0/2	0/2	3.004	0.465
High single R	+36	U170 $\alpha = 0$	2/2	0/2	0/2	2.476	0.465
High single R	+36	U170 $\alpha = 1$	2/2	0/2	2/2	2.769	0.503
Low S R	-36	Rule	2/2	0/2	0/2	1.946	1.446
Low S R	-36	U170 $\alpha = 0$	1/2	0/2	0/2	1.750	2.060
Low S R	-36	U170 $\alpha = 1$	1/2	0/2	0/2	1.875	1.976
Low S R	+36	Rule	1/2	0/2	0/2	1.972	1.000
Low S R	+36	U170 $\alpha = 0$	2/2	1/2	0/2	1.640	0.368
Low S R	+36	U170 $\alpha = 1$	2/2	2/2	2/2	1.807	0.273
Low single L	-36	Rule	2/2	0/2	0/2	1.853	1.964
Low single L	-36	U170 $\alpha = 0$	0/2	0/2	0/2	1.563	3.083
Low single L	-36	U170 $\alpha = 1$	0/2	0/2	0/2	1.744	2.163
Low single L	+36	Rule	2/2	0/2	0/2	1.853	1.260
Low single L	+36	U170 $\alpha = 0$	0/2	0/2	0/2	1.558	2.545
Low single L	+36	U170 $\alpha = 1$	0/2	0/2	0/2	1.745	2.253

Table 4: Individual matched cases in the illustrated +36 N positive subset.

Route	Bank	Method	Goal	Wide	Speed	Max cross	Return RMS
Low S R	2	Rule	0	0	1.969	1.406	0.872
Low S R	2	U170 $\alpha = 0$	1	0	1.638	0.476	0.140
Low S R	2	U170 $\alpha = 1$	1	1	1.789	0.268	0.103
Low S R	5	Rule	1	0	1.975	0.595	0.638
Low S R	5	U170 $\alpha = 0$	1	0	1.642	0.261	0.114
Low S R	5	U170 $\alpha = 1$	1	1	1.824	0.278	0.095
High single R	2	Rule	1	0	3.009	0.466	1.199
High single R	2	U170 $\alpha = 0$	1	0	2.470	0.440	0.133
High single R	2	U170 $\alpha = 1$	1	1	2.761	0.502	0.062
High single R	5	Rule	1	0	3.000	0.464	1.276
High single R	5	U170 $\alpha = 0$	1	0	2.482	0.490	0.134
High single R	5	U170 $\alpha = 1$	1	1	2.776	0.503	0.048

4 Physical-domain and prepared-state transfer

The two domain models were selected from an existing four-draw TEST set by the maximum and minimum wheel-friction values. Domain 0 used wheel friction 0.932 and chassis mass scale 0.975; domain 3 used 0.612 and 1.074. Each domain used its own complete prepared banks 0 and 6. Table 5 gives every domain/route/method grouping. All 48 physical episodes reached the original destination and survived H10. The new task protocol was applied after earlier outcomes were inspected, so these data show behavior in two sampled domains rather than an untouched generalization estimate.

Table 5: Task transfer by physical domain, original route and method.

Domain	Route	Method	Narrow	Wide	Speed	Max cross
0	High S L	Rule	2/2	2/2	3.054	0.112
0	High S L	U170 $\alpha = 0$	0/2	2/2	2.692	0.454
0	High S L	U170 $\alpha = 1$	0/2	2/2	2.880	0.707
0	High single R	Rule	2/2	2/2	3.012	0.108
0	High single R	U170 $\alpha = 0$	2/2	0/2	2.512	0.209
0	High single R	U170 $\alpha = 1$	0/2	2/2	2.805	0.858
0	Low S R	Rule	1/2	1/2	1.957	0.456
0	Low S R	U170 $\alpha = 0$	2/2	0/2	1.614	0.217
0	Low S R	U170 $\alpha = 1$	2/2	2/2	1.764	0.279
0	Low single L	Rule	1/2	1/2	1.847	0.299
0	Low single L	U170 $\alpha = 0$	2/2	0/2	1.559	0.216
0	Low single L	U170 $\alpha = 1$	2/2	2/2	1.738	0.278
3	High S L	Rule	2/2	2/2	3.044	0.104
3	High S L	U170 $\alpha = 0$	0/2	2/2	2.680	0.445
3	High S L	U170 $\alpha = 1$	0/2	2/2	2.870	0.705
3	High single R	Rule	2/2	2/2	3.004	0.107
3	High single R	U170 $\alpha = 0$	2/2	0/2	2.490	0.229
3	High single R	U170 $\alpha = 1$	0/2	2/2	2.780	0.837
3	Low S R	Rule	2/2	2/2	1.955	0.168
3	Low S R	U170 $\alpha = 0$	2/2	0/2	1.639	0.225
3	Low S R	U170 $\alpha = 1$	2/2	2/2	1.806	0.276
3	Low single L	Rule	1/2	1/2	1.823	0.648
3	Low single L	U170 $\alpha = 0$	2/2	1/2	1.647	0.204

Domain	Route	Method	Narrow	Wide	Speed	Max cross
3	Low single L	U170 $\alpha = 1$	2/2	2/2	1.879	0.278

5 Smooth 90-degree turn experiment

The added finite-radius original routes change heading by exactly $\pm 90^\circ$ at the end of each turn; the source NPZ paths were independently checked. Low-speed requests use 2.0 m/s and radius 1.6 m; high-speed requests use 3.05 m/s and radius 3.0 m, each in left and right directions. Every method used the same two prepared states, original destination and task thresholds. All 48 episodes entered the original goal and survived H10. Table 6 gives the complete route/preference grouping. The rule completed both task scores in 8/8 contexts; U170 exhibits a consistent preference response while not exceeding that rule on these turns. Figure 2 shows the right-turn trajectories omitted from the main article's preselected left-turn figure.

Table 6: All smooth 90° route/preference groupings.

Route	Method	Narrow	Wide	Speed	Max cross
High 90 L	Rule	2/2	2/2	2.996	0.113
High 90 L	U170 $\alpha = 0$	0/2	2/2	2.758	0.622
High 90 L	U170 $\alpha = 0.25$	0/2	2/2	2.823	0.701
High 90 L	U170 $\alpha = 0.5$	0/2	2/2	2.883	0.769
High 90 L	U170 $\alpha = 0.75$	0/2	2/2	2.935	0.827
High 90 L	U170 $\alpha = 1$	0/2	2/2	2.973	0.881
High 90 R	Rule	2/2	2/2	3.010	0.112
High 90 R	U170 $\alpha = 0$	2/2	0/2	2.509	0.214
High 90 R	U170 $\alpha = 0.25$	2/2	0/2	2.537	0.340
High 90 R	U170 $\alpha = 0.5$	0/2	1/2	2.592	0.523
High 90 R	U170 $\alpha = 0.75$	0/2	2/2	2.683	0.679
High 90 R	U170 $\alpha = 1$	0/2	2/2	2.793	0.810
Low 90 L	Rule	2/2	2/2	1.953	0.150
Low 90 L	U170 $\alpha = 0$	2/2	0/2	1.633	0.273
Low 90 L	U170 $\alpha = 0.25$	2/2	0/2	1.686	0.298
Low 90 L	U170 $\alpha = 0.5$	2/2	2/2	1.750	0.330
Low 90 L	U170 $\alpha = 0.75$	2/2	2/2	1.811	0.372
Low 90 L	U170 $\alpha = 1$	0/2	2/2	1.867	0.431
Low 90 R	Rule	2/2	2/2	1.968	0.120
Low 90 R	U170 $\alpha = 0$	2/2	0/2	1.588	0.192
Low 90 R	U170 $\alpha = 0.25$	2/2	0/2	1.608	0.187
Low 90 R	U170 $\alpha = 0.5$	2/2	0/2	1.635	0.187
Low 90 R	U170 $\alpha = 0.75$	2/2	0/2	1.678	0.228
Low 90 R	U170 $\alpha = 1$	2/2	2/2	1.750	0.285

Complete case and raw-trajectory access

The machine-readable companions retain every individual outcome, including non-arrivals and task failures: nominal cases, signed pulses, physical domains, and 90-degree turns. Their adjacent Markdown indices link all original/actual XY images and archived raw NPZ trajectories. The current paper package has no new policy training and no hardware results.

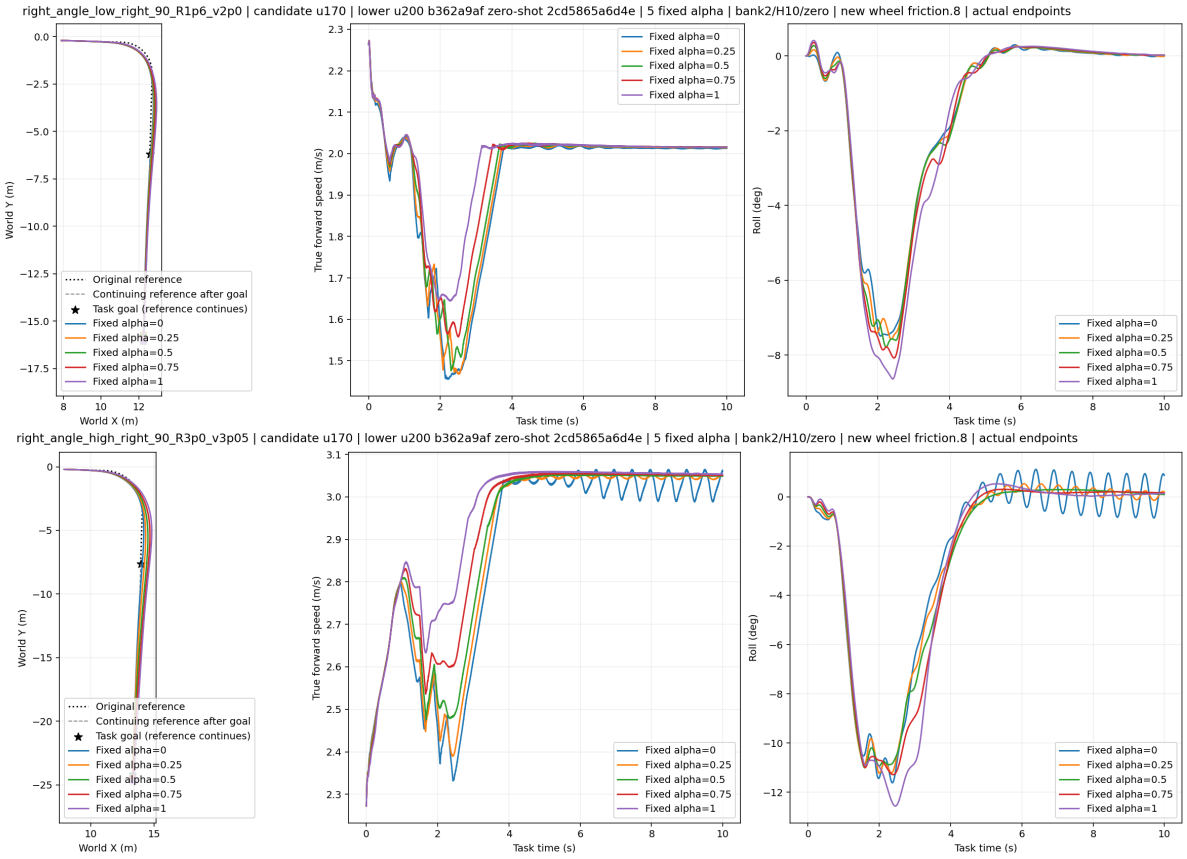


Figure 2: Right 90° turns at low and high requested speed, prepared bank 2: all five U170 preferences, original path, actual speed and roll. The corresponding rule and bank-5 trajectories are in the full result index.

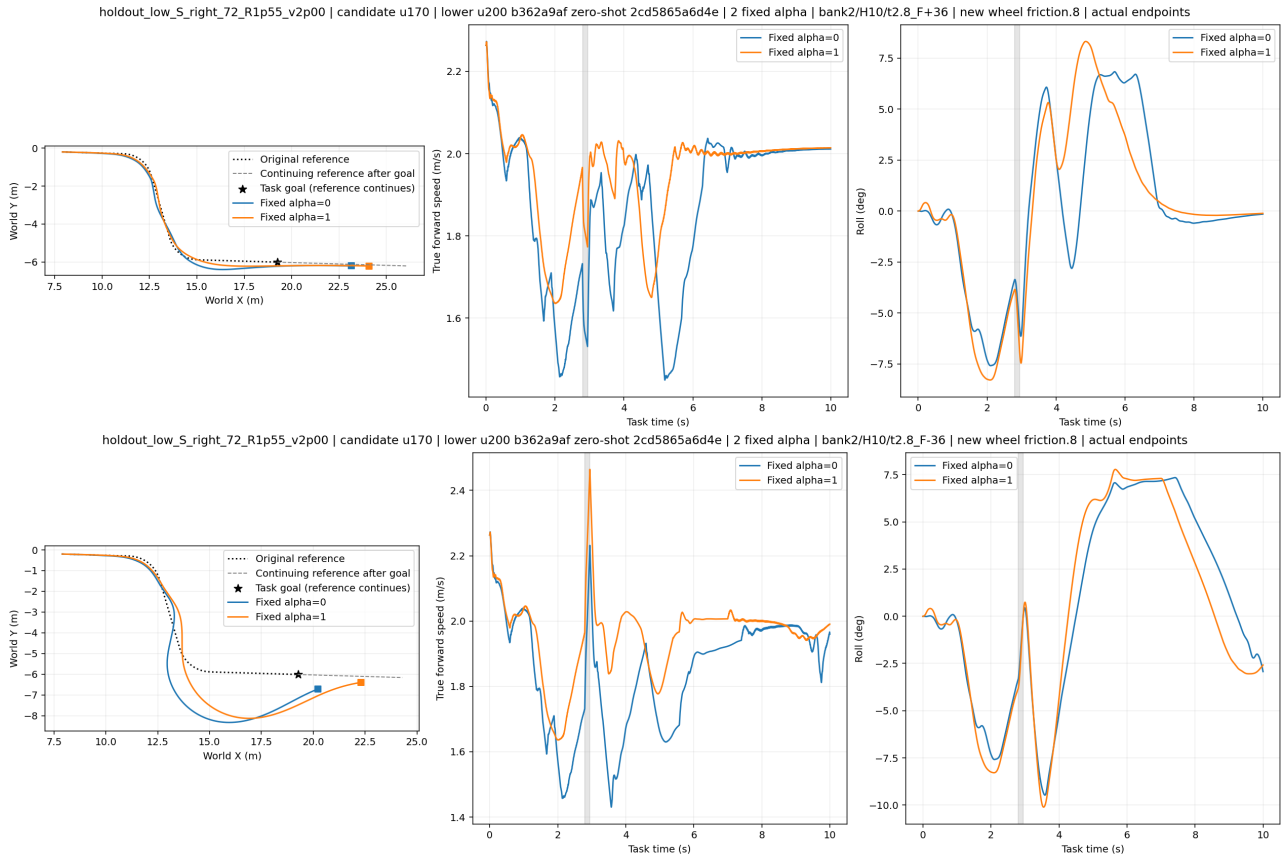


Figure 3: Same low-speed S right route and prepared bank 2 under the two signed force pulses. The article's positive example uses the upper +36 N case; the lower −36 N case is retained for direction-specific comparison.